

Design Stage

We identified the entities, activities, and agents required for our project based on a chronological thought process, developed by examining the work we performed each week. Initially, only one agent was included (Team Magenta) but we later added two more (Team Pink and Green) due to the indicator we developed using other teams' metadata. The activities and agents were chosen based on their relevance to the project, as we opted for mid-level granularity. This was done following the provenance convention. For example, for the activity Survey Creation, Qualtrics was used, and so included that as the entity.

Throughout the modelling, our approach remained consistent. We started with the responsibility view, then considered the data flow view, and lastly the process view. Types were decided based on what an entity or agent represented. For example, all teams included groups of individuals with shared goals; hence, the type "Group". Similarly, identifiers were named accurately and specifically for easy understanding. For example, `ex:SurveyCreation` is clearer than `ex:CreateActivity` because it specifies *what* is being created. We made sure to use Pascal-case throughout to enhance readability and consistency. Entities like the software used to create the indicator and the electronic devices used to fill the surveys were not included.

The survey respondents were not included as agents. This is because these entities and agents were not imperative for the design process of our indicator, and so we felt it would add too much noise to the diagram.

Considering Granularity

Suppose we had modelled our provenance for lower granularity. In that case, there may be fewer informative labels, such as the number of respondents and start/end times of activities performed. Details such as exterior agent contributions and the activities they performed to create the metadata entities, on which we based our employability indicator, could be omitted to simplify the indicator design process. For instance, the activity of 'metadata creation' for teams Pink and Green could be grouped into a single activity, rather than describing each team's actions. On the other hand, our provenance could be more detailed for researchers interviewing the same subjects by including version numbers and modification timestamps to show how our entities changed over time. Our roles within Team Magenta could be elaborated to incorporate our main contributions. In addition, we could break down the process of our employability indicator creation into separate tasks, such as deciding which metadata to include and standardising data on different scales.

Reflection

While the coding aspect of the provenance was not particularly challenging for the group, considerations of the relevant entities to include in the diagram proved more complex. The formalisation process of our provenance required recalling dates we had not recorded, which may have led to a lack of precision in the diagram. For future reference, building the provenance up gradually during the process could improve the clarity of the diagram. On top of that, consideration of the granularity when attributing agents to the activities resulted in

some difficult decisions. As seen in the final diagram, we decided to do this on a group level, with the individual personal agents being omitted and a joint responsibility given for each task, as we mostly worked collaboratively. In general, the group worked together well and we were largely in agreement on the actions we took.

PROV-N Diagram



